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Advanced Expiry Risk, Waste Analytics, and Reorder Optimization Dashboard for HimShakti Food Processing Unit

1. BUSINESS PROBLEM STATEMENT

HimShakti Food Processing Unit currently faces severe profit margin erosion due to an absolute lack of data visibility across its operational pipeline. Specifically, there is no system to track raw material batch expiry against the finished goods production schedules. This data gap leads to high volumes of unmonitored material waste, operational inefficiencies, and dead stock. This project addresses the challenge by constructing an automated, predictive inventory health dashboard that monitors material shelf life, calculates ongoing financial losses, and flags critical expiration timelines to directly secure and stabilize business margins.

2. TARGET AUDIENCE

The primary stakeholders and users of this analytical dashboard include:

- **Founder:** Requires high-level visibility into financial leakages, overall operational health scores, and margin protection trends.
- **Operations Manager:** Needs real-time data to align batch production schedules with the actual shelf life of raw materials in stock.
- **Procurement Lead:** Relies on automated reorder alerts and vendor-specific shelf-life quality metrics to optimize buying decisions.

3. DATASET IDENTIFIED & ARCHITECTURE

For proxy data modeling, the project utilizes an industry-standard open-source dataset.

Dataset Name	Source / URL	Volume	Key Modeled Columns
Food Industry Inventory & Supply Chain Dataset	Kaggle Proxy Data kaggle.com/datasets	500+ Rows	`Item_Name`, `Batch_Number`, `Received_Date`, `Expiry_Date`, `Quantity_Received`, `Supplier_ID`

4. KEY BUSINESS QUESTIONS ADDRESSED

#	Core Analytical Inquiry	Operational Value & Impact
Q1	What is the total financial loss (INR) incurred from expired inventory over a rolling 30-day window?	Quantifies explicit revenue leakage to motivate corrective financial interventions.
Q2	Which core raw materials fall under the critical threshold of the Reorder Alert System (Current Stock < Reorder Level)?	Triggers dynamic operational reordering to eliminate stockouts without bloating warehouse capital.
Q3	Which suppliers consistently deliver batch ingredients with the lowest average remaining shelf life?	Provides empirical scorecard data to rewrite supplier SLA contracts and reduce intake risk.

5. PROPOSED TOOL & TECHNOLOGY STACK

- **Data Pipeline:** `Python (pandas)` will be utilized for ingestion, missing value imputation, date parsing, and localized data transforming logic.
- **Visualization Layer:** `Matplotlib` and `Seaborn` will render operational visual charts, breakdown graphs, and threshold indicators.
- **GenAI Layer:** `Gemini API` integration will synthesize rule-based operational flags into contextual, natural-language procurement recommendations.

6. EXPECTED CAPSTONE OUTPUTS & DELIVERABLES

The successful execution of this project will yield the following operational assets:

- **Executive Analytical Report:** A robust synthesis outlining procurement and production strategy modifications to mitigate stock expiration.
- **High-Level Inventory Health Scorecard:** A unified composite metric evaluating immediate stock health, risk distribution, and waste velocity.
- **Working BI Dashboard:** A functional analytical interface featuring automated reorder triggers and proactive aging alerts.
- **Localized SOP Document:** A standardized operational playbook ensuring sustained process tracking and dashboard continuity for the HimShakti team.